

Quiz 1: Engineering Mechanics (Dynamics)

Dynamics of single particle

Total Questions: 14

Total Marks: 135

Total time: 3 hours (10 AM to 1 PM)

Bring your own scientific calculator.

If you complete quiz early, submit your answer sheet and leave the room quietly.

If you are found using any unfair means, you will be asked to submit the answer sheet and **you should leave the room immediately. Do not argue.** Your answer sheet will not be evaluated.

1. Derive expressions for velocity and acceleration in normal-tangential coordinates.

Hints:

- a) Start from the expression of velocity (direction of velocity vector is along the tangent to the path)
- b) Use the concept of radius of curvature
- c) Positive director of normal unit vector is towards the center of curvature.

Marks:10

2. The speed of a car increases uniformly with time from 50 km/h at A to 100 km/h at B during 10 seconds. The radius of curvature of the hump at A is 40 m. If the magnitude of the total acceleration of the mass center of the car is the same at B as at A , compute the radius of curvature of the dip in the road at B . The mass center of the car is 0.6 m from the road.

Marks:10

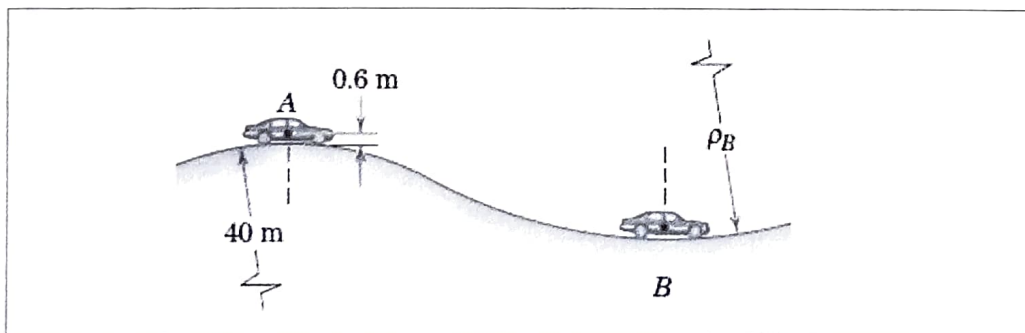


Fig. 1

3. Derive expressions for velocity and acceleration in polar coordinates (r-theta coordinates).

Hints:

- Start from expression of position vector in polar coordinates
- Derive expressions of rate of change of unit vectors
- Derive expression of velocity as rate of change of position vector
- Derive expression of acceleration as rate of change of velocity vector

Marks:10

4. An internal mechanism is used to maintain a constant angular rate $\Omega = 0.05 \text{ rad/s}$ about the z-axis of the spacecraft as the telescopic booms are extended at a constant rate. The length l is varied from essentially zero to 3 m. The maximum acceleration to which the sensitive experiment modules P may be subjected is 0.011 m/s^2 . Determine the maximum allowable boom extension rate $\dot{l}$.

Marks:10

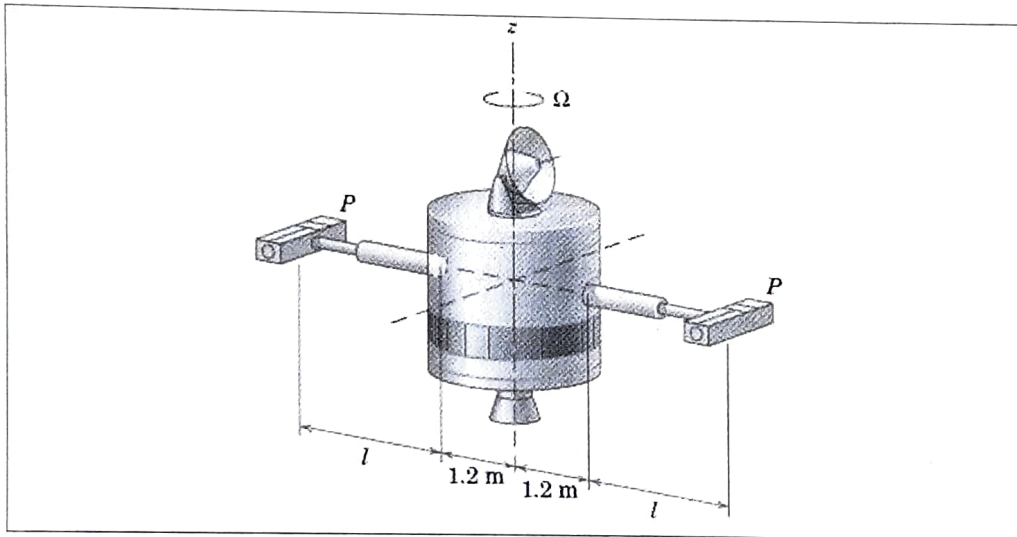


Fig. 2

5. An aircraft is flying in a horizontal circle of radius b with a constant speed u at an altitude of h . A radar tracking unit is located at C. Write expressions for the components of the velocity of the aircraft in the spherical coordinates of the radar station for a given position β .

Marks: 10

6. An amusement ride called the “corkscrew” takes the passengers through the upside-down curve of a horizontal cylindrical helix. The velocity of the cars as they pass position A is 15 m/s, and the component of their acceleration measured along the tangent to the path is $g \cos \gamma$ at this point. The effective radius of the cylindrical helix is 5 m, and the helix angle is $\gamma = 40^\circ$. Compute the magnitude of the acceleration of the passengers as they pass position A.

Marks: 10

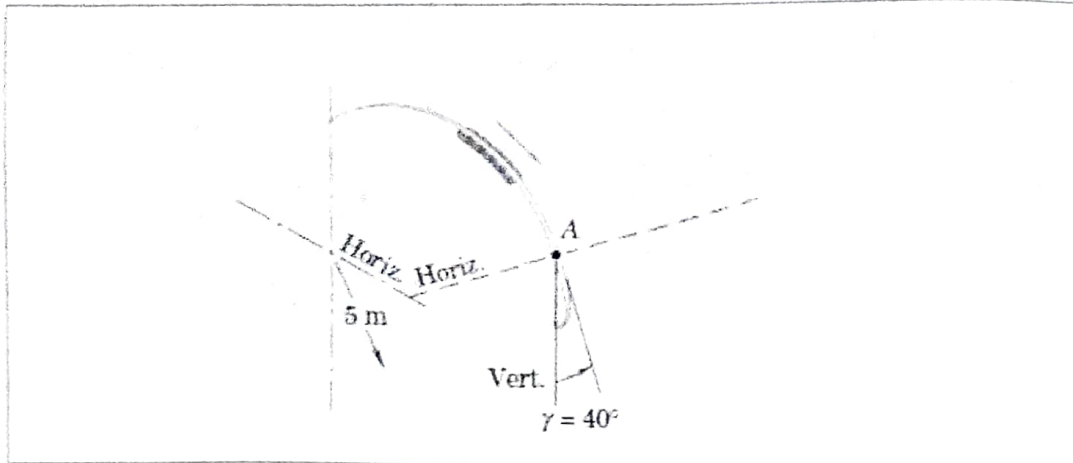


Fig. 4

7. Prove that work energy principle is applicable when measurements are made with respect to a frame of reference moving with constant velocity.
NOTE: It is written 'Frame of reference moving with constant velocity'

Marks: 10

8. The system is released from rest with no slack in the cable and with the spring stretched 200 mm. Determine the distance s traveled by the 10-kg cart before it comes to rest (a) if m approaches zero and (b) if $m = 2 \text{ kg}$. Assume no mechanical interference.

Marks: 10

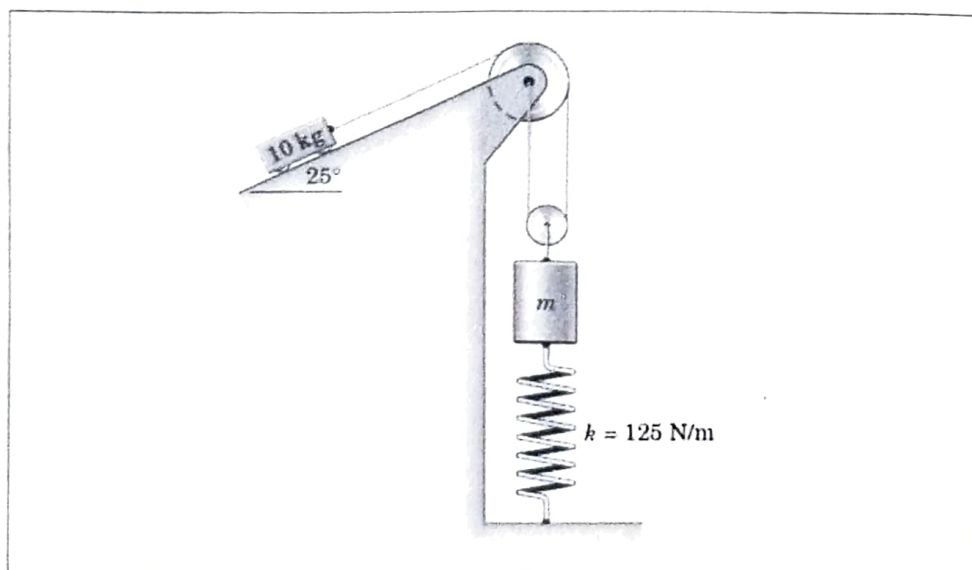


Fig. 5

9. The cars of an amusement-park ride have a speed $v_1 = 90 \text{ km/hr}$ at the lowest part of the track. Determine their speed v_2 at the highest part of the track. Neglect energy loss due to friction. (Caution: Give careful thought to the change in potential energy of the system of cars.)

Marks: 10

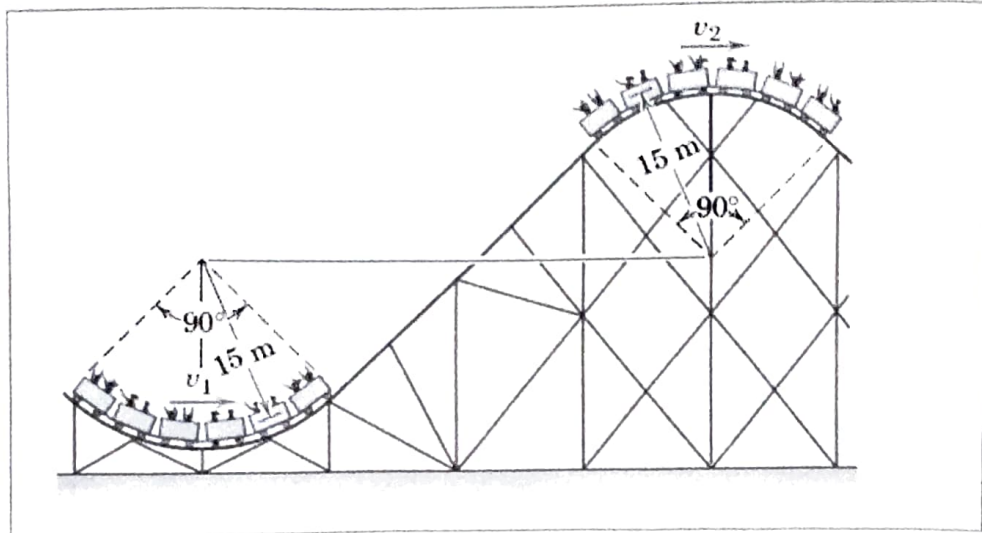


Fig. 6

10. Prove that the angular impulse - angular momentum principle is applicable when measurements are made with respect to a frame of reference moving with constant velocity.

Hint: Angular momentum is a moment of linear momentum. The question is about frame of reference moving with constant velocity.

Marks: 10

11. A particle is launched with a horizontal velocity $v_0 = 0.55 \text{ m/s}$ from the 30° position shown and then slides without friction along the funnel-like surface. Determine the angle θ which its velocity vector makes with the horizontal as the particle passes level $O-O$. The value of r is 0.9 m .

Marks: 10

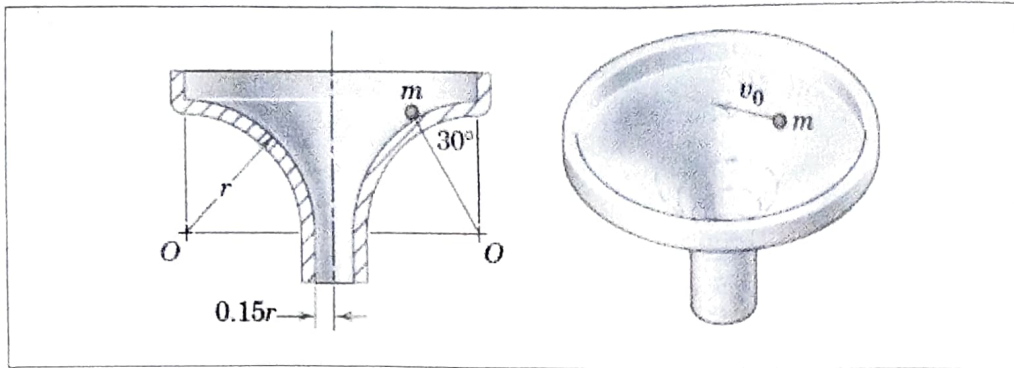


Fig. 7

12. Write, explain, and derive Kepler's third law of planetary motion.

Marks:10

13. The launch catapult of the aircraft carrier gives the 7-Mg jet airplane a constant acceleration and launches the airplane in a distance of 100 m measured along the angled takeoff ramp. The carrier is moving at a steady speed $v_c = 16$ m/s. If an absolute aircraft speed of 90 m/s is desired for takeoff, determine the net force F supplied by the catapult and the aircraft engines.

Marks:10

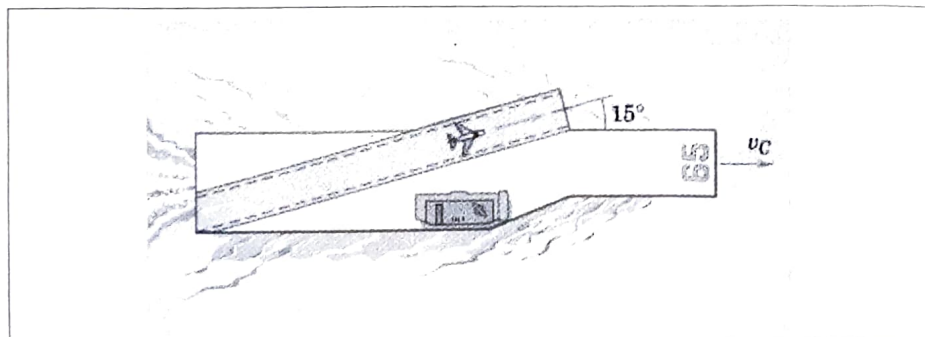


Fig. 8

14. Explain the concept utilized to solve problem no. 13.

Marks: 5

Quiz 2: Engineering Mechanics (Dynamics)
Dynamics of System of Particles and Dynamics of Rigid Bodies

Total Questions: 18

Total Marks: 165 (=150+15)

Total time: 3 hours (10 AM to 1 PM)

Bring your own scientific calculator. If you complete quiz early, submit your answer sheet and leave the room quietly.

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1. Derive the expression for absolute velocity and absolute acceleration of point A in terms of absolute acceleration of point B and measurements made in the rotating frame of reference (x-y frame of reference) [Refer to Fig. 1].

Marks: 10

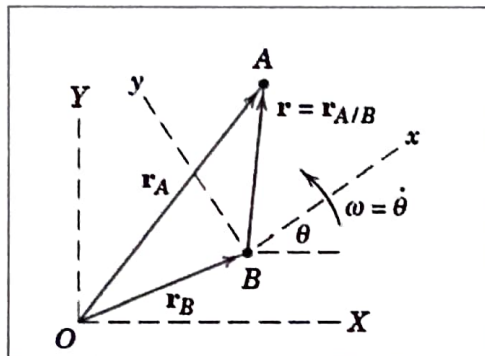


Fig. 1 – Fixed frame of reference (X-Y) and rotating frame of reference (x-y)

2. The circular disk of radius, $r = 0.6$ m rotates about a fixed axis through point O with the angular properties, $\omega = 2$ rad/s and $\alpha = 3$ rad/s² with directions as shown in the figure. Determine the instantaneous values of the velocity and acceleration of point A.

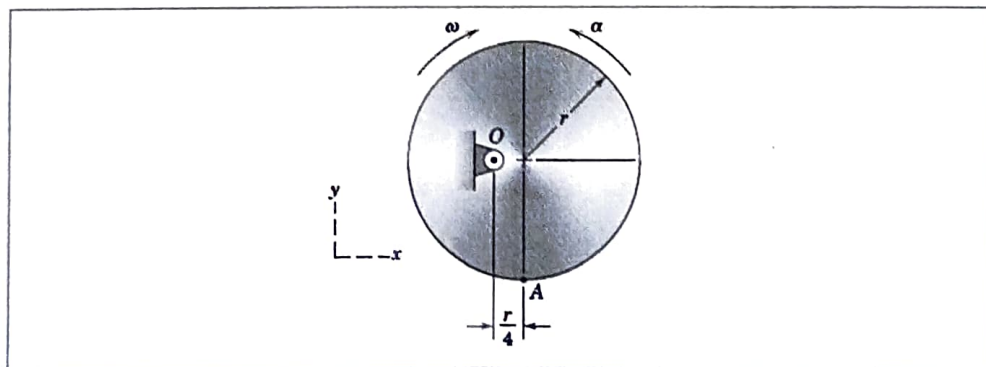


Fig. 2

Marks: 10

3. Explain the physical meaning of Coriolis acceleration for planar motion. You may refer to Fig. 3. Also mention mathematical expression for Coriolis acceleration.

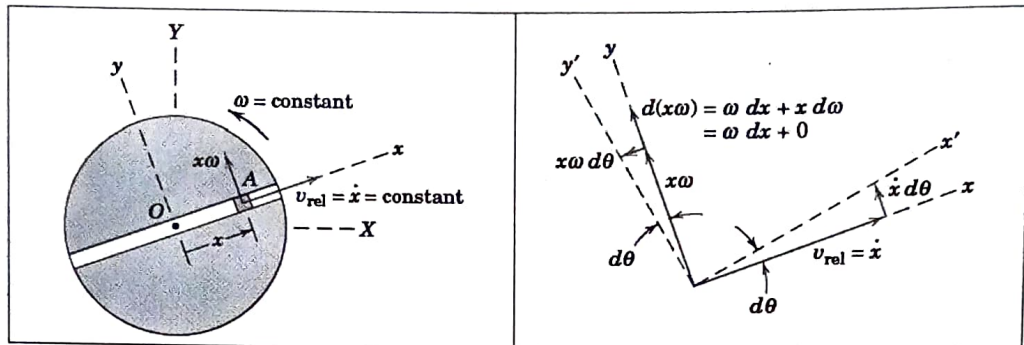


Fig. 3

Marks: 10

4. The disk rotates about a fixed axis through O with angular velocity, $\omega = 5 \text{ rad/s}$ and angular acceleration, $\alpha = 3 \text{ rad/s}^2$ in the directions shown at a certain instant. The small sphere A moves in the circular slot, and at the same instant, $\beta = 30^\circ$, $\dot{\beta} = 2 \text{ rad/s}$ and $\ddot{\beta} = -4 \text{ rad/s}^2$. Determine the absolute velocity and acceleration of A at this instant.

Marks: 10

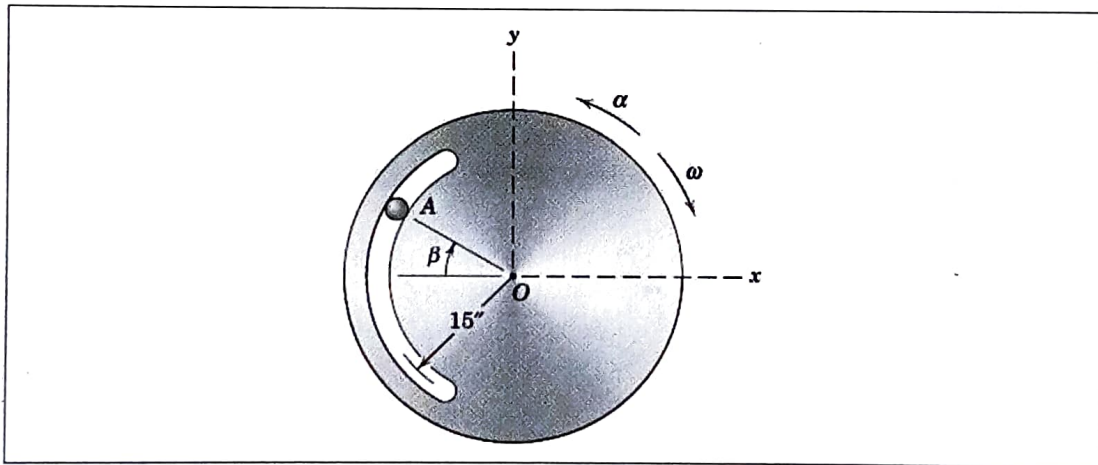


Fig. 4

5. The wheel slips as it rolls. If $v_0 = 4 \text{ ft/s}$ and if the velocity of A with respect to B is $3\sqrt{2} \text{ ft/s}$, locate the instantaneous center C of zero velocity and find the velocity of point P.

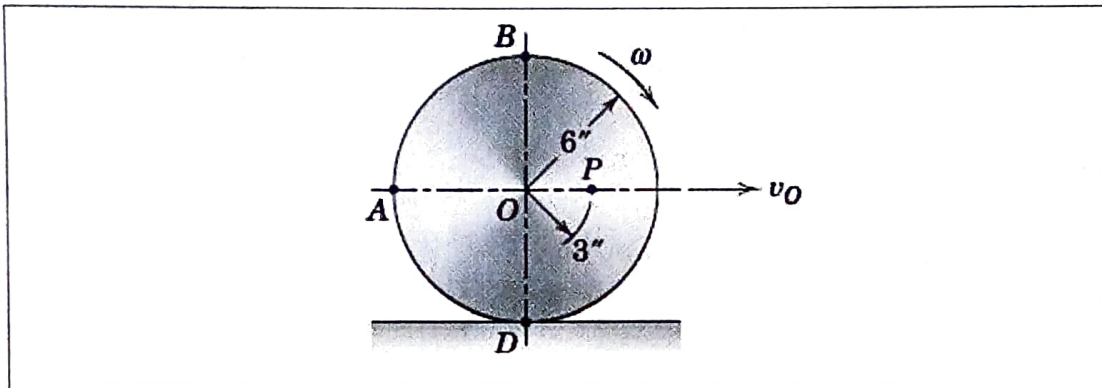


Fig. 5

Marks: 10

6. Derive Newton's second law of motion of a system of n particles.

Hint: 1. Write Newton's second law of motion of one particle considering external and internal forces. 2. Sum expression obtained in step 1 for all the n particles.

Marks: 5

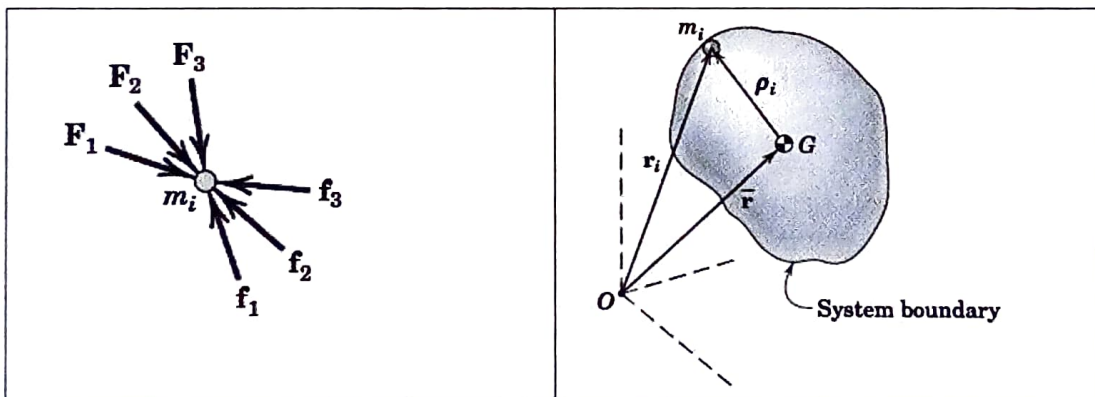


Fig. 6

7. The three small spheres are connected by the cords and spring and are supported by a smooth horizontal surface. If a force $F = 6.4 \text{ N}$ is applied to one of the cords, find the acceleration of the mass center of the spheres for the instant depicted.

Marks:5

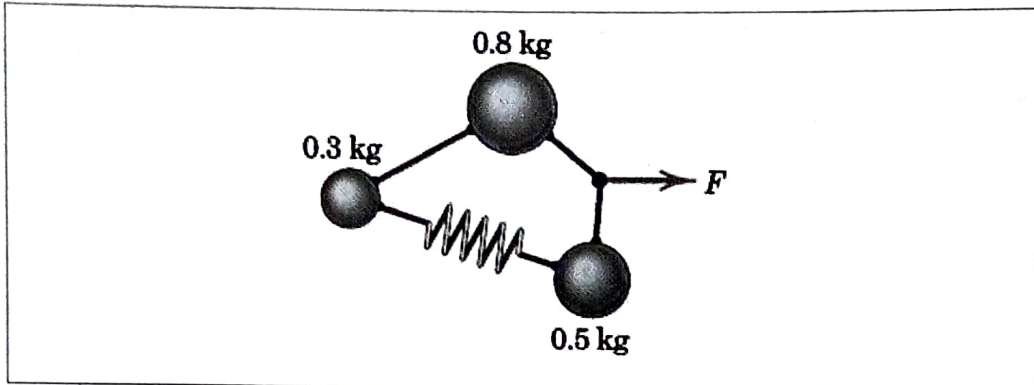


Fig. 7

8. Show that for a system of n particles, total kinetic energy of the system can be written in terms of velocity of center of mass and relative velocities of all the particles with respect to center of mass.

Marks:10

$$T = \frac{1}{2} (m\bar{v}^2 + \sum m_i |\dot{\rho}_i|^2)$$

9. What do you understand by 'absolute angular momentum' and 'relative angular momentum'?

Marks:5

10. The two small spheres, each of mass m , are rigidly connected by a rod of negligible mass and are released from rest in the position shown and slide down the smooth circular guide in the vertical plane. Determine their common velocity v as they reach the horizontal dashed position. Also find the force R between sphere 1 and the supporting surface an instant before the sphere reaches the bottom position A .

Marks:10

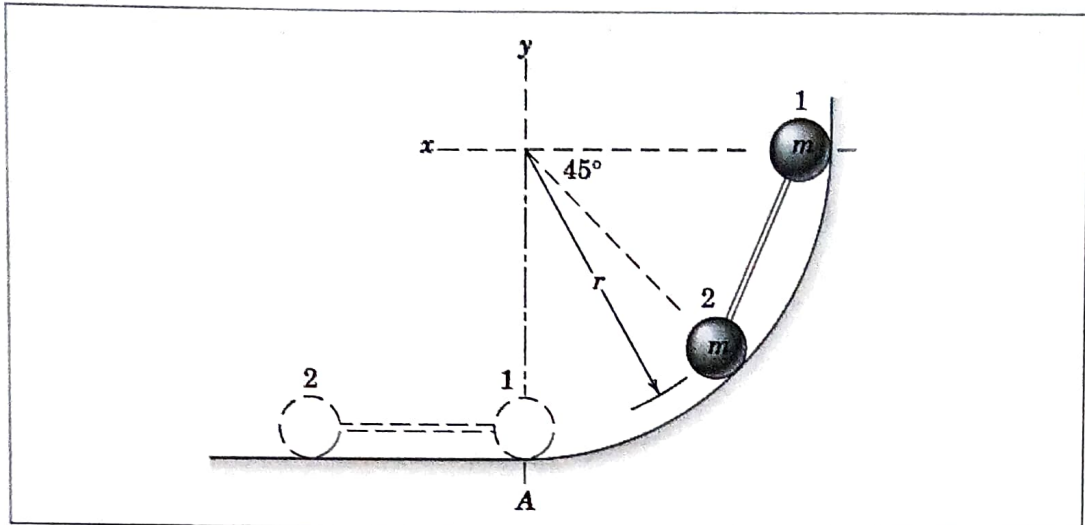


Fig. 8

11. Show that for a system of n particles, absolute angular momentum about center of mass is same as its relative angular momentum about center of mass. [Refer to Fig. 9]

Marks:10

$$(\mathbf{H}_G)_{rel} = (\mathbf{H}_G)_{absolute}$$

12. Derive expression, for a system of n particles, relating rate of change of angular momentum about origin of a fixed reference system and moment of all the forces acting on all the particles about the same point. Do moments by internal forces need to be considered while calculating the total moment of all the forces about the fixed point? [Refer to Fig. 9]

Marks:10

$$\dot{H}_o = \sum M_o$$

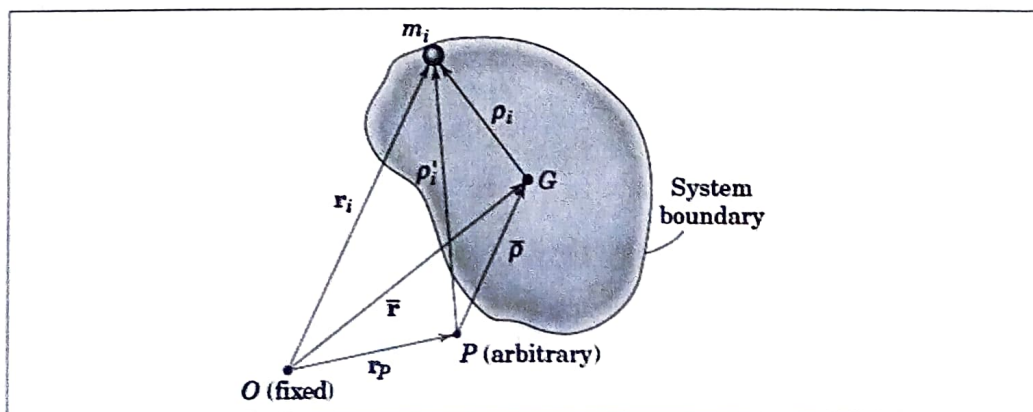


Fig. 9

13. Show that for a system of n particles, rate of change of angular momentum about center of mass is equal to moment of all the forces acting on all the particles about center of mass. Do moments by internal forces need to be considered while calculating the total moment of all the forces about the center of mass? [Refer to Fig. 9]

Marks:10

$$\dot{H}_G = \sum M_G$$

14. Show that for a system of n particles, absolute angular momentum about any arbitrary point P equals the angular momentum about G (center of mass) plus the moment about P of the linear momentum of the system considered concentrated at G . [Refer to Fig. 9]

$$H_P = H_G + \bar{\rho} \times m\bar{v}$$

Marks:10

15. Show that for a system of n particles, rate of change of angular momentum about an arbitrary point P is not equal to moment of all the forces acting on all the particles about the same point. [Refer to Fig. 10]

Marks:10

$$H_P = H_G + \bar{\rho} \times m\bar{v}$$

$$\sum M_P = \dot{H}_G + \bar{\rho} \times m\bar{a}$$

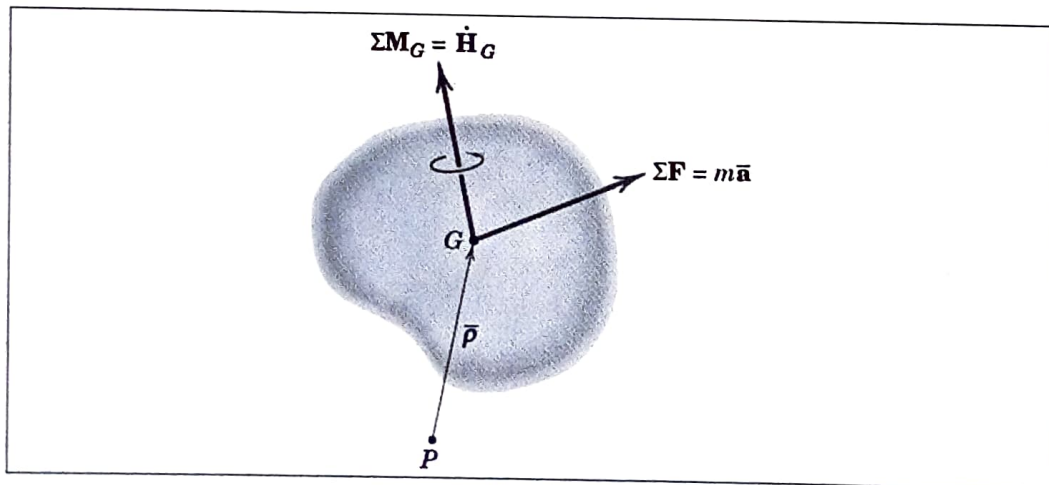


Fig. 10

16. Small ball bearing-rollers mounted on the ends of the slender bar of mass m and length l constrain the motion of the bar in the horizontal $x - y$ slots. If a couple M is applied to the bar initially at rest at $\theta = 45^\circ$, determine the forces exerted on the rollers at A and B as the bar starts to move.

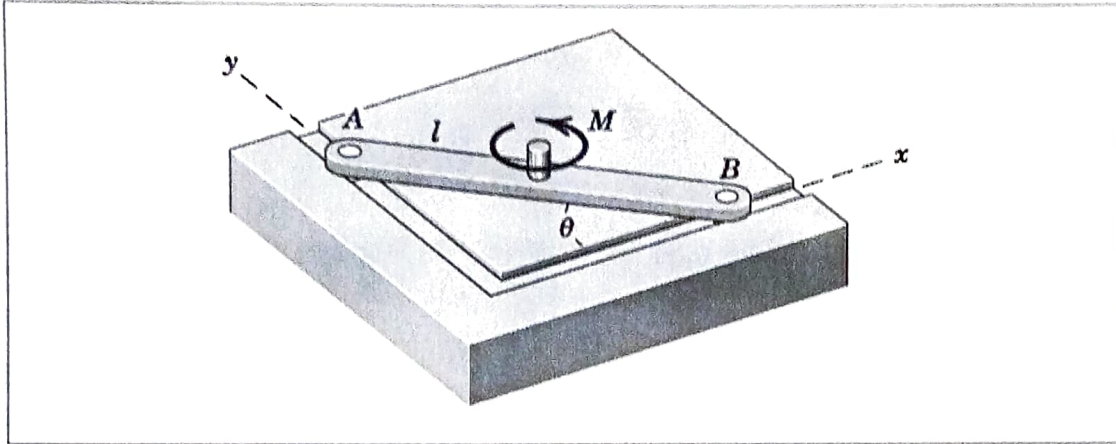


Fig. 11

Marks:10

17. Derive the expression, for a system of n particles, between moment of all the forces about any arbitrary point P and rate of change of relative angular momentum about P . [Refer to Fig. 9]

$$\sum \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{rel} + \bar{\boldsymbol{\rho}} \times m \mathbf{a}_p$$

If

- (a) $\bar{\boldsymbol{\rho}} = 0$
- (b) $\mathbf{a}_p = 0$
- (c) $\bar{\boldsymbol{\rho}}$ and $\mathbf{a}_p$ are parallel

$$\sum \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{rel}$$

Marks:10

18. The small car, which has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\dot{\theta} = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0^\circ$, calculate v when $\theta = 60^\circ$. Neglect the mass of the wheels and any friction.

Marks: 10

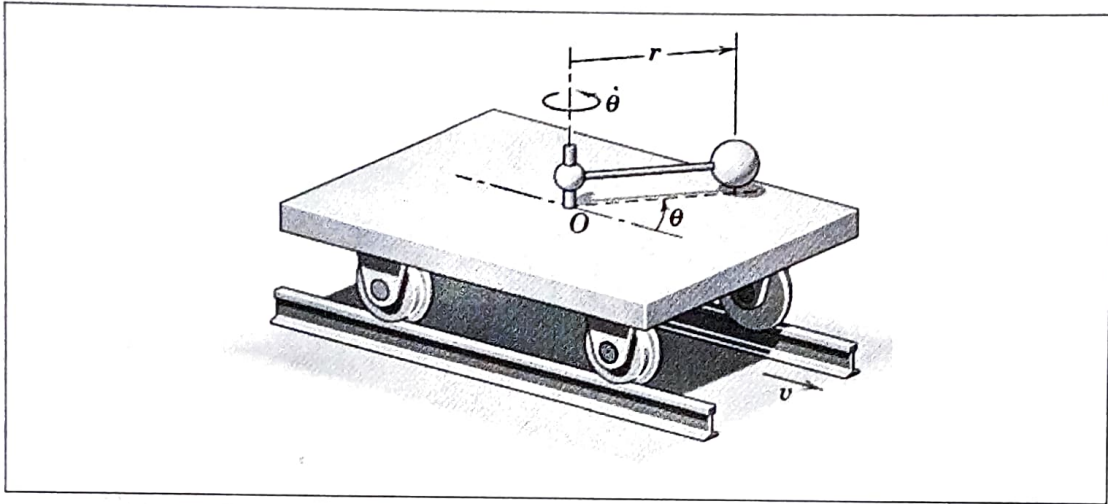


Fig. 12

Quiz 3: Engineering Mechanics (Dynamics)

Dynamics of Rigid Bodies

Total Questions: 10

Total Marks: 15+10+10+10+10+10+15+10+10+10=110

Total time: 3 hours (10 AM to 1 PM)

Bring your own scientific calculator. If you complete quiz early, submit your answer sheet and leave the room quietly. If you are found using any unfair means, you will be asked to submit the answer sheet and **you should leave the room immediately. Do not argue.** Your answer sheet will not be evaluated.

- 1)
 - a. Derive expression for angular momentum for a rigid body in 3-D motion.
 - b. Explain inertia tensor. What are products of inertia and moments of inertia?
 - c. What are principal axes of inertia? What do you understand by principal moments of inertia?
 - d. What are principal curvatures of a curved surface? [If you do not know, Google it after the quiz]
 - e. Write expression for angular momentum when axes (x-y-z) aligns with principal axes of inertia?
 - f. What is 'center of percussion'? Explain significance of center of percussion.
 - g. Write force and moment equations for pure translation, pure rotation and general plane motion of a rigid body.

Marks: 12+3=15

- 2) What is Gyroscope? What do you understand by term 'precession'?
Derive momentum equations for Gyroscopic motion? From momentum equations, derive Euler's momentum equations for gyroscopic motion? [Refer to Fig. 1 and Fig. 2]

Marks: 10

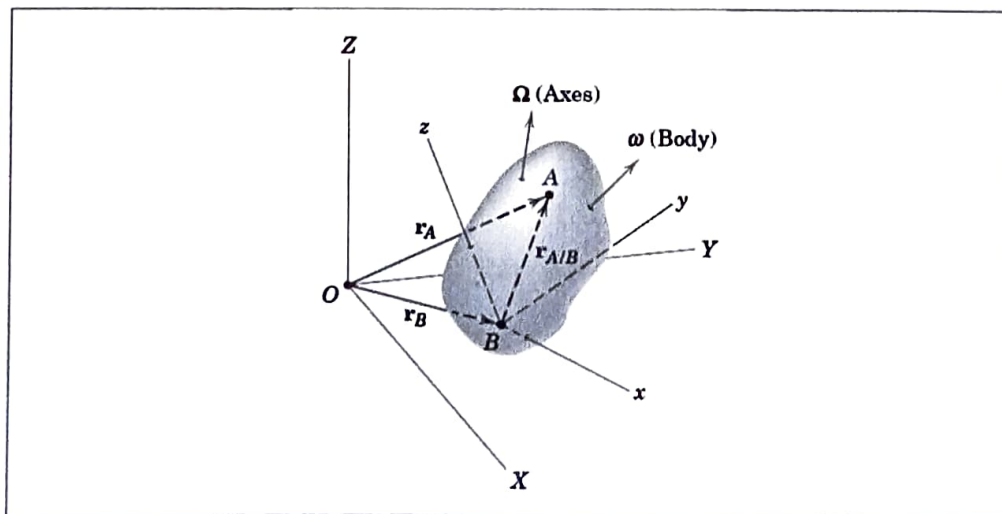


Fig. 1

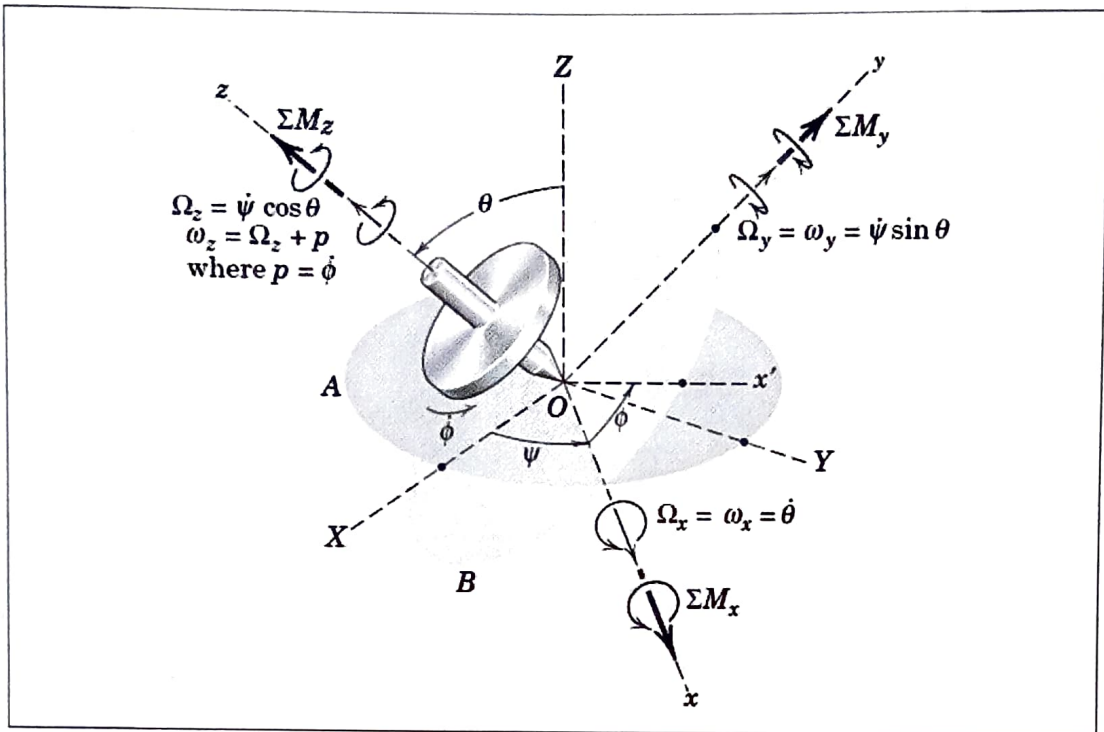


Fig. 2

- 3) An experimental vehicle A travels with constant speed v relative to the earth along a north-south track.
- Determine the Coriolis acceleration $\mathbf{a}_{cor}$ as a function of the latitude θ . Assume an earth-fixed rotating frame $Bxyz$ and a spherical earth.
 - If the vehicle speed is $v = 500 \text{ km/hr}$, determine the magnitude of the Coriolis acceleration at (a) the equator and (b) the north pole.
- Hint: One day = 24 hours = 24×60 minutes = $24 \times 60 \times 60$ seconds

Marks: 10

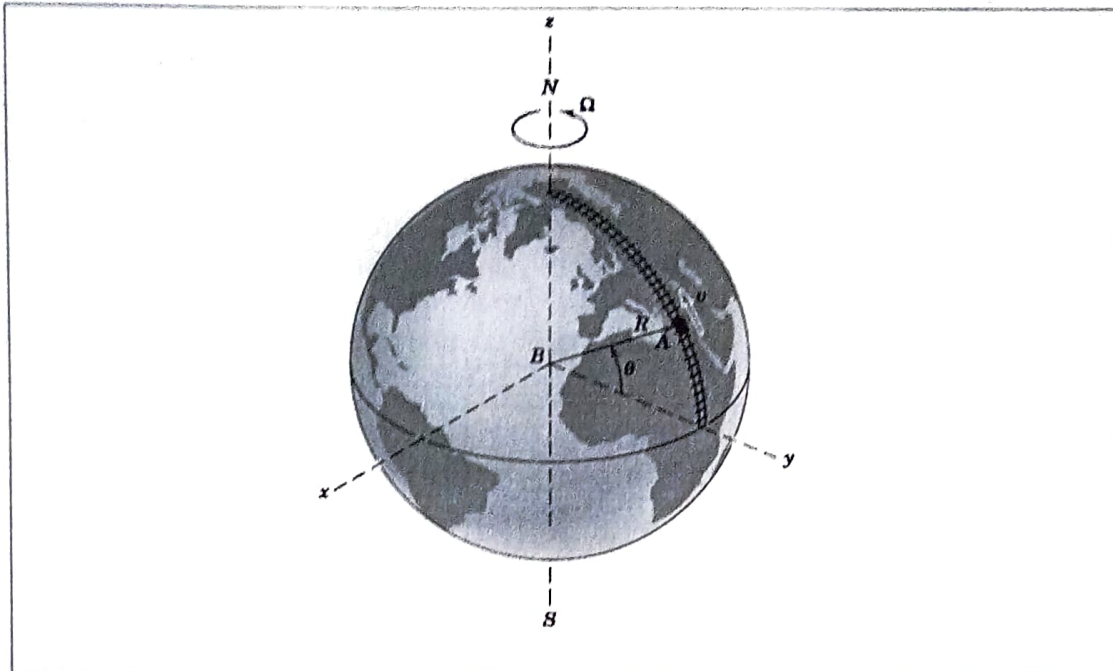


Fig. 3

- 4) Two small variable-thrust jets are actuated to keep the spacecraft angular velocity about the z-axis constant at $\omega_0 = 1.25 \text{ rad/s}$ as the two telescoping booms are extended from $r_1 = 1.2 \text{ m}$ to $r_2 = 4.5 \text{ m}$ at a constant rate over a 2-min period. Determine the necessary thrust T for each jet as a function of time where $t = 0$ is the time when the telescoping action is begun. The small 10 kg experiment modules at the ends of the booms may be treated as particles, and the mass of the rigid booms is negligible.

Marks: 10

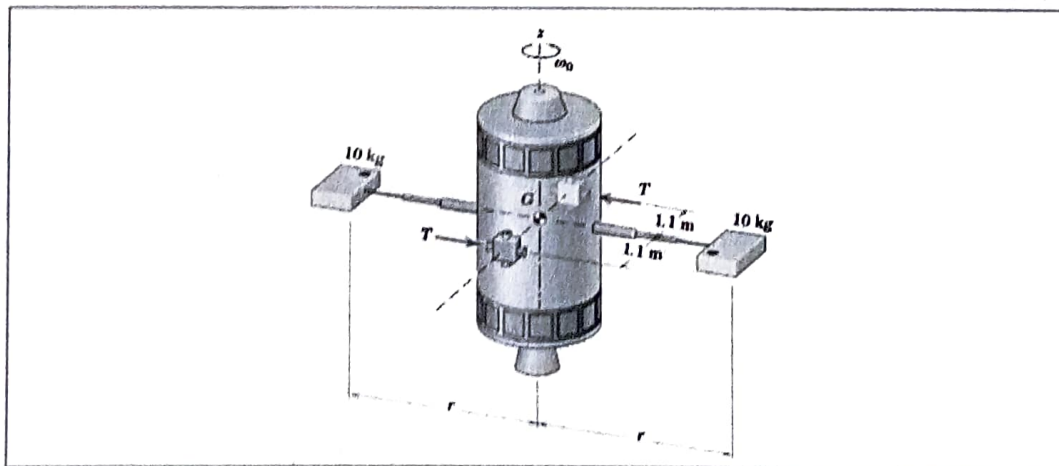


Fig. 4

- 5) The uniform 200-kg bar AB is raised in the vertical plane by the application of a constant couple $M = 3 \text{ kN}\cdot\text{m}$ applied to the link at C . The mass of the links is small and may be neglected. If the bar starts from rest at $\theta = 0$, determine the magnitude of the force supported by the pin at A as the position $\theta = 60^\circ$ is passed.

Hint: Think whether this is curvilinear translation or rotation of the body about fixed axis or general plane motion. Apply corresponding force and moment equation about center of mass and about any other point.

Marks: 10

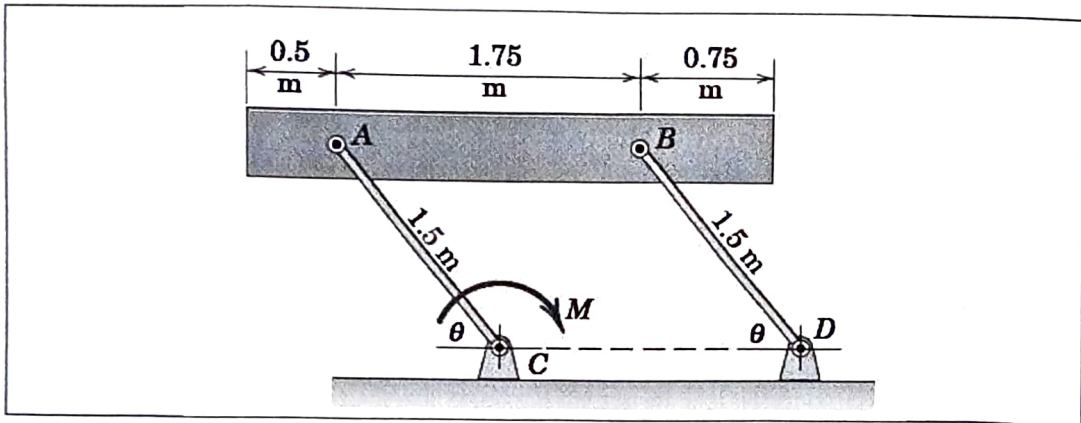


Fig. 5

- 6) The wheel consists of a 4-kg rim of 250-mm radius with hub and spokes of negligible mass. The wheel is mounted on the 3-kg yoke OA with mass center at G and with a radius of gyration about O of 350 mm. If the assembly is released from rest in the horizontal position shown and if the wheel rolls on the circular surface without slipping, compute the velocity of point A when it reaches at A' .

Hint: Work-Energy Principle

Marks: 15

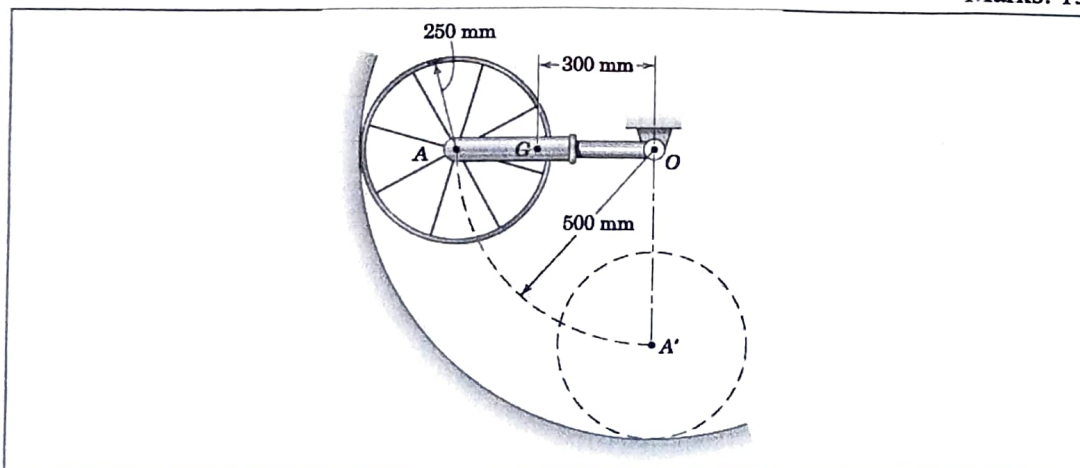


Fig. 6

- 7) The spacecraft shown has a mass m with mass center G . Its radius of gyration about its z -axis of rotational symmetry is k and that about either the x - or y -axis is k' . In space, the spacecraft spins within its x - y - z reference frame at the rate $p = \dot{\phi}$. Simultaneously, point C on the z -axis moves in a circle about the z_0 -axis with a frequency f (rotations per unit time). The z_0 -axis has a constant direction in space. Determine the angular momentum $\mathbf{H}_G$ of the spacecraft relative to the axes designated. Note that the x -axis always lies in the z - z_0 plane and that the y -axis is therefore normal to z .

Hint: Angular momentum can be expressed easily about principal axes of inertia. Spin velocity is p . $H = I_{xx}\omega_x + I_{yy}\omega_y + I_{zz}\omega_z$

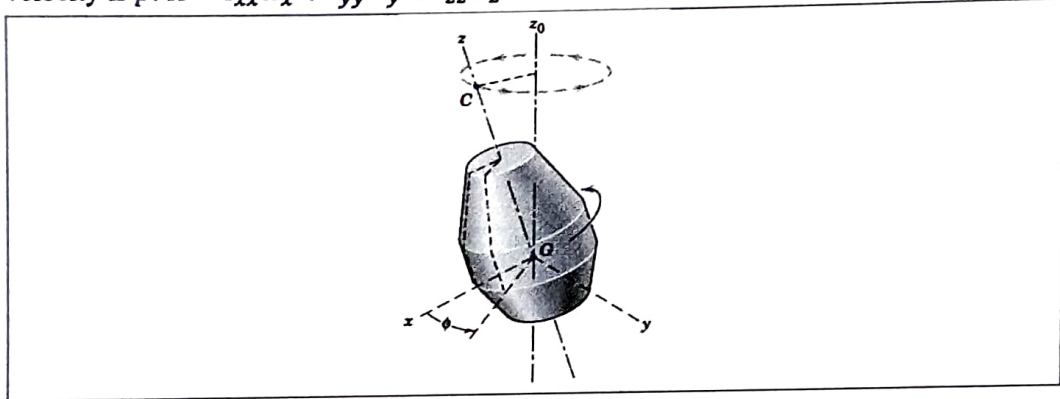


Fig. 7

Marks:10

- 8) The 8 kg rotor with radius of gyration of 3 cm rotates on ball bearings at a speed of 3000 rev/min about its shaft OG. The shaft is free to pivot about the X -axis, as well as to rotate about the Z -axis. Calculate the vector $\mathbf{\Omega}$ for precession about the Z -axis. Neglect the mass of shaft OG and compute the gyroscopic couple $\mathbf{M}$ exerted by the shaft on the rotor at G .

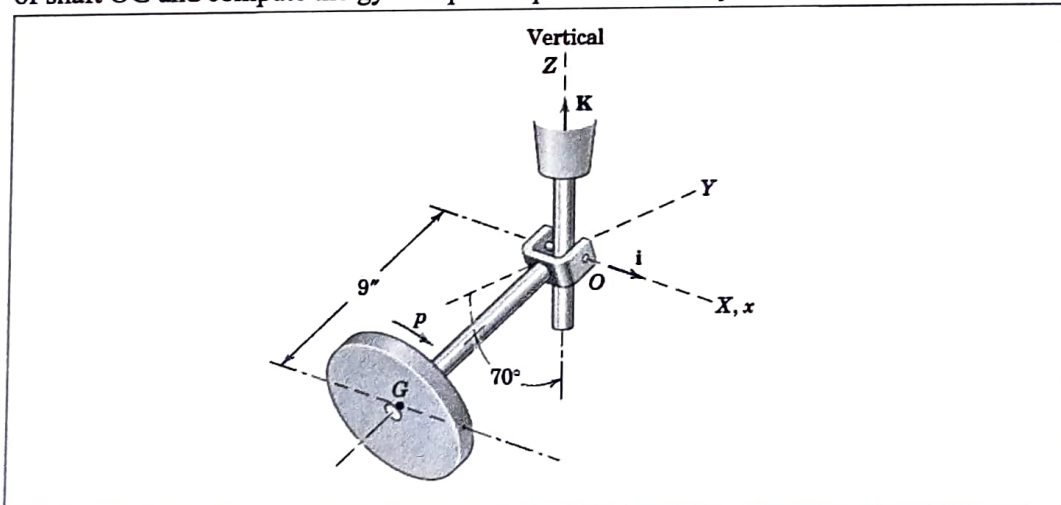


Fig. 8

Marks: 10

- 9) If the motor pivots about the x -axis at the constant rate $\dot{\gamma} = 3\pi$ rad/sec with no rotation about the Z -axis ($N=0$), determine the angular acceleration α of the rotor and disk as the position $\gamma = 30^\circ$ is passed. The constant speed of the motor is 120 rev/min. Also find the velocity and acceleration of point A , which is on the top of the disk for this position. $OC=10$ cm and $CA=5$ cm.

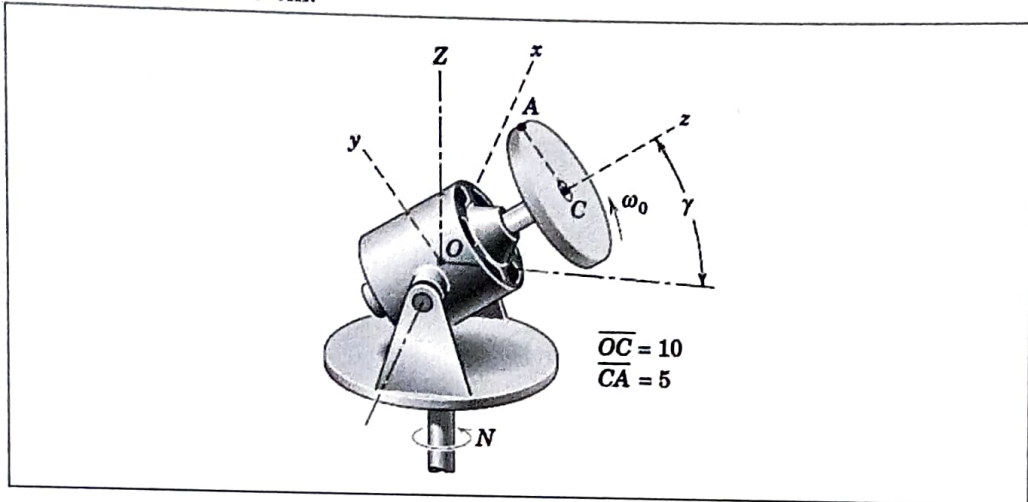


Fig. 9

Marks:10

After the exam: How to proceed in Q. 12 if N (rotation about Z -axis) has some constant value (not zero)?

- 10) A uniform circular disk which rolls with a velocity v without slipping encounters an abrupt change in the direction of its motion as it rolls onto the incline θ . Determine the new velocity v' of the center of the disk as it starts up the incline, and find the fraction n of the initial energy which is lost because of impact with the incline if $\theta = 10^\circ$.

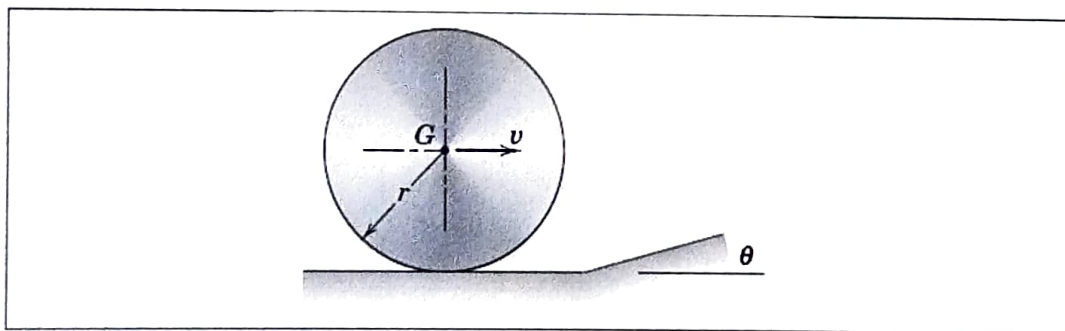


Fig. 10

Marks: 10

Engineering Mechanics MEL 1010
Dynamics of particles Quiz-5

Total Time: 2 hrs [Exam time] + 30 minutes (extra time)
Total Number of Questions: 20

Total Marks: 126

If you complete the quiz early, submit your answer sheet and leave the examination room quietly.

1. Explain
 - a) Instantaneous center of zero velocity
 - b) Body centre and space centre
 - c) Significance of point of contact or tangent point between body centre and space centre
 - d) Instantaneous axis of rotation
 - e) Space cone and body cone
 - f) Significance of contact line or tangent line between body cone and space cone

Marks: 12

2. Derive expression for rate of change of unit vectors (along x-axis and y-axis) of rotating frame of reference (x-y) rotating with angular velocity of ω as shown in Fig. 1.

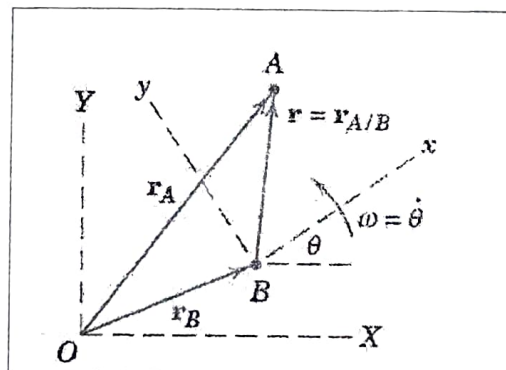


Fig. 1 – Fixed frame of reference (X-Y) and rotating frame of reference (x-y)

Marks: 6

3. Derive the expression for absolute velocity of point A in terms of absolute velocity of point B and measurements made in rotating frame of reference (x-y frame of reference) [Refer to Fig. 1]. Explain the physical meaning of all the terms.

Marks: 6

4. Derive the expression for absolute acceleration of point A in terms of absolute acceleration of point B and measurements made in the rotating frame of reference (x-y frame of reference) [Refer to Fig. 1]. Explain the physical meaning of all the terms.

Marks: 10

~~$r \times m \omega$~~
 $(r \times m \dot{r})$

5. Explain the physical meaning of Coriolis acceleration for planar motion. You may refer to Fig. 2. Also mention mathematical expression for Coriolis acceleration.

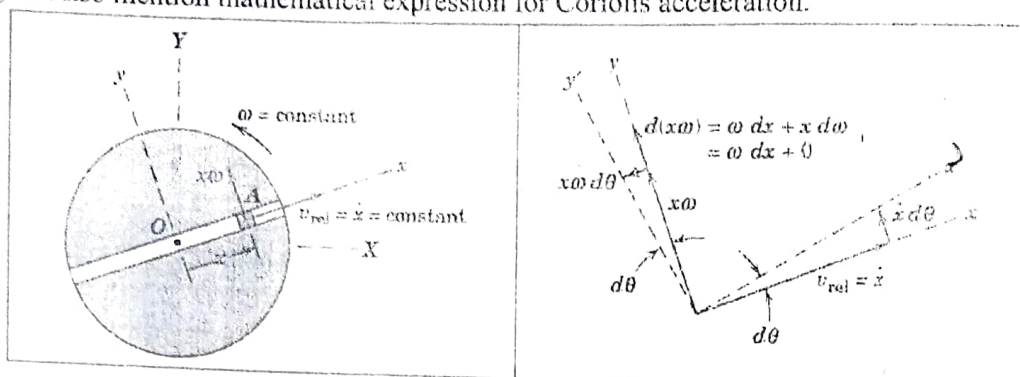


Fig. 2

Marks: 5

6. Show that for a planar motion,
- The angular momentum of a rigid body about the center of mass can be written as product of mass moment of inertia about center of mass and angular velocity of the body.
 - The angular momentum of a rigid body about any arbitrary point O can be written as angular momentum of the body about center of mass and moment of linear momentum considering that total mass is concentrated at center of mass and velocity of center of mass.

Marks: 6

- 7.
- For a rigid body undergoing curvilinear translation, write moment equation.
 - Write force equation for motion of a rigid body rotating about a fixed axis passing through its center of mass, G .
 - Write moment equation for a rigid body undergoing fixed axis rotation about a point O .
 - Write force equation and moment equation for a rigid body undergoing general plane motion.

Marks: 4

8. What is the center of percussion? Explain the significance of the center of percussion in context of a tennis racket or Cricket bat.

Marks: 4

9. Write expression of kinetic energy of a rigid body undergoing a planar motion
- rotating about a fixed axis of rotation (expression of KE in terms of mass moment of inertia and angular velocity)
 - undergoing general plane motion (expression of KE in terms of translational velocity of the center of mass and angular velocity about center of mass)

Marks: 4

10. With reference to Fig. 3, write an expression for rate of change of unit vectors along coordinate axes of rotating frame of reference for 3D motion. ω is the angular velocity of the rigid body and Ω is the angular velocity of rotating frame of reference.

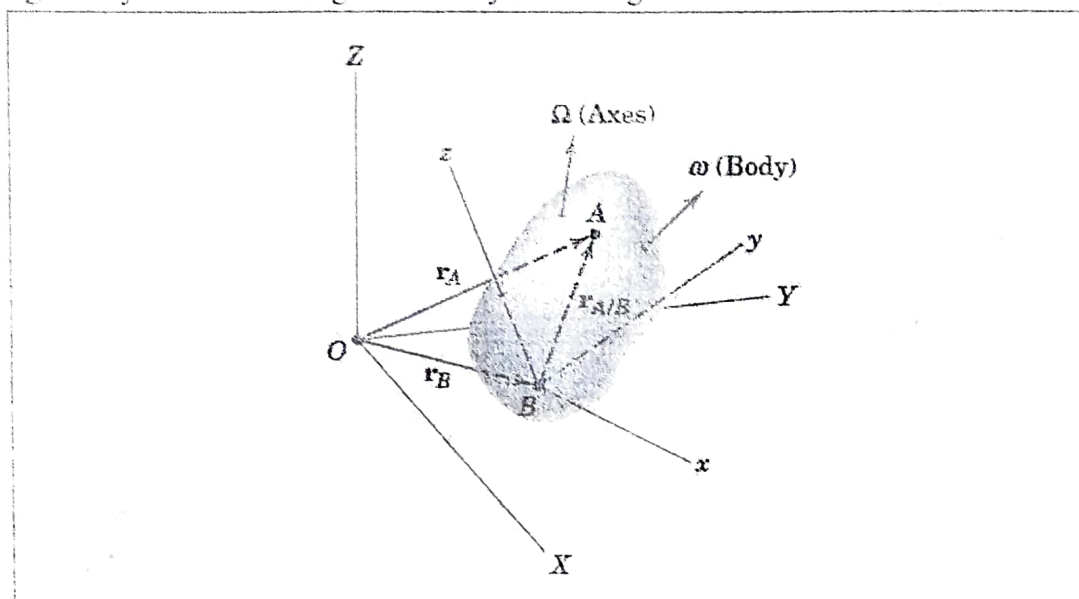


Fig. 3

Marks: 3

11. Write the expression for absolute velocity of A and absolute acceleration of A when measurements are made with respect to rotating frame of reference $x-y-z$. [Refer to Fig. 3]. Give an example of Coriolis acceleration observed in 3D motion.

Marks: 6

12. Explain

- Products of inertia and moments of inertia
- Principal axes of inertia and principal moments of inertia

Marks: 4

13. Derive the expression for **angular momentum** of a rigid body in terms of products of inertia and moments of inertia when the axes of frame of reference are fixed to the rigid body and hence, rotating with the same angular velocity as that of the body. You can assume that the origin of the frame of reference is at the center of mass of the body. Draw relevant figures.

Marks: 8

14. Write the expression for angular momentum of a rigid body when frame of reference coincides with principal axes of inertia of the rigid body.

Marks: 4

$$(x^2 + y^2 + z^2)(\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}) - (\omega_x x + \omega_y y + \omega_z z)(x \hat{i} + y \hat{j} + z \hat{k})$$

15. Explain transfer theorem of angular momentum. You may refer to Fig. 4.

$$\begin{pmatrix} y\omega_x - x\omega_y \\ z\omega_x - x\omega_z \\ x\omega_y - y\omega_z \end{pmatrix} = \begin{pmatrix} y\omega_x - x\omega_y \\ z\omega_x - x\omega_z \\ x\omega_y - y\omega_z \end{pmatrix} = \begin{pmatrix} y\omega_x - x\omega_y \\ z\omega_x - x\omega_z \\ x\omega_y - y\omega_z \end{pmatrix} = \begin{pmatrix} y\omega_x - x\omega_y \\ z\omega_x - x\omega_z \\ x\omega_y - y\omega_z \end{pmatrix}$$

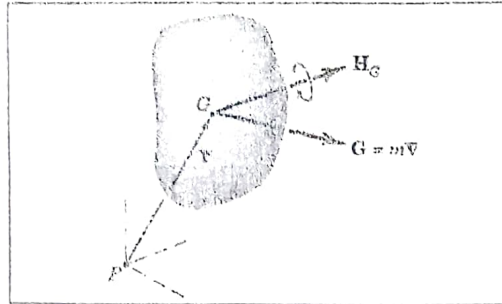


Fig. 4

Marks: 4

16. Show that kinetic energy of a rigid body can be written in terms of angular momentum about the center of mass and linear momentum of the body. The linear momentum is computed as product of total mass (assumed to be concentrated at center of mass) and velocity of center of mass.

Marks: 6

17. Derive Euler equations for motion of a rigid body. **Hint:** Start from momentum equations. The angular velocity of rotating frame of reference can be assumed to be same as the angular velocity of the rigid body.

Marks: 10

18. Explain Fig. 5 and write momentum equations for the same.

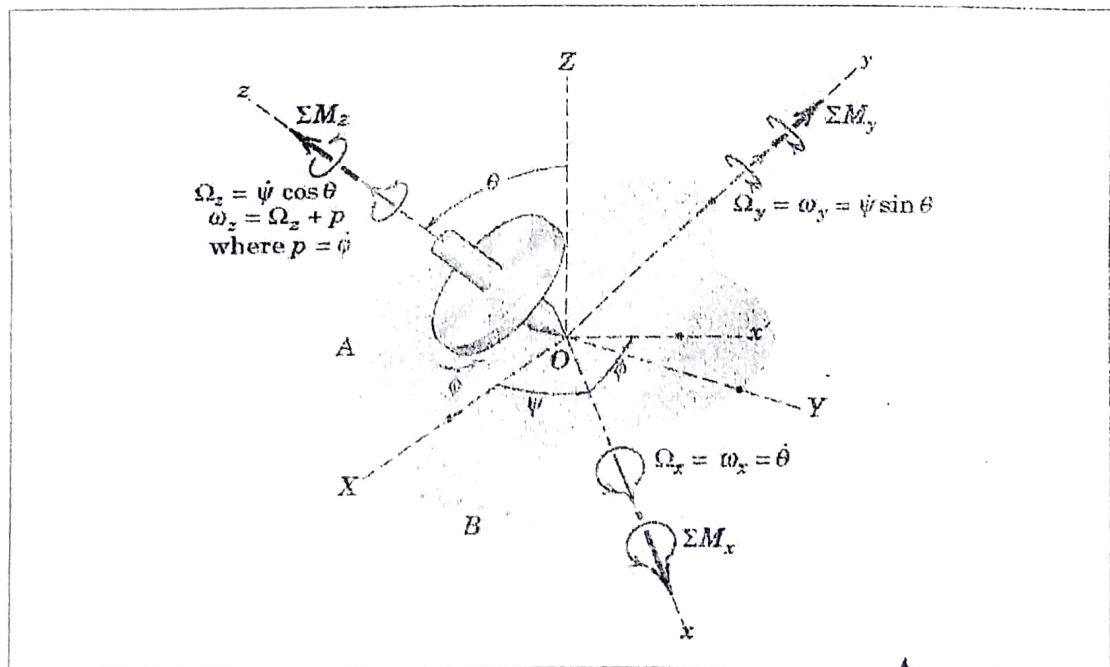


Fig. 5

Marks: 10

19. Write momentum equations for a rotor system which is rotating about its spin axis and undergoing a steady state precession about another axis (refer to Fig. 5). **Hint:** Write conditions of steady state precession.

Marks: 8

20. From Fig. 6, identify the case of direct precession and retrograde precession. Explain your answer (reasoning behind the answer). It is to be observed that for both the cases, no force acts on rotating body which is also undergoing precession about another axis. **Hint:** Observe moment of inertia of the rotating body about y-axis and z-axis.

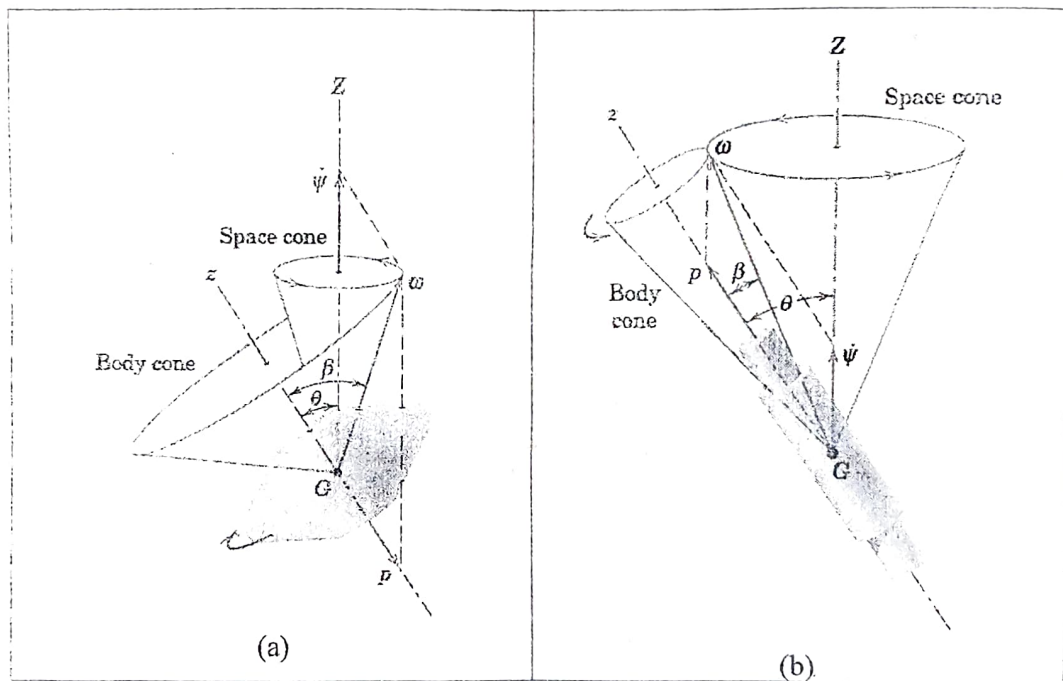


Fig. 6

Marks: 6